

Report: training the ActiveLearning project

Date: September 25, 2026 · **Machine:** RTX 5070, 12 GB · **Data:** 1,660 training plates, 415 validation

What was added, because it was missing

- 1. Quality metrics.** Training only reported the loss, which can go down without the model detecting anything. Precision, recall and mAP are now computed at the end of every epoch.
- 2. The DINOv3 backbone.** The project shipped a four-layer feature extractor with no pretraining, marked as a placeholder. `MODEL_STRATEGY.md` called for DINOv3, but it was never implemented. It was added.
- 3. The RT-DETR detector.** This is experiment 1 as documented by the author, and it did not exist in the code. It was added as a new file, without touching the original detector.
- 4. Two fixes needed to run at all:** an untranslated label (`Listeria_otros_bact`) that made the LM head impossible to convert; and the PyTorch version, which made everything run on CPU when the GPU was required.

Results

By experiment (values from each run's best epoch):

Run	Backbone	Frozen?	Detector	Batch	How it differs from what the author specified	mAP	Prec.	Recall	Duration
Baseline (original)	simple	no	9-head DETR	16	Uses the placeholder backbone, not DINOv3	0.00011	7.99 %	0.38 %	77 min
Experiment 2	DINOv3	YES	9-head DETR	16	Backbone frozen	0.00012	0.23 %	0.24 %	125 min
Experiment 1	DINOv3	YES	RT-DETR	8	Backbone frozen and batch reduced	0.00055	0.79 %	1.55 %	144 min
Exp. 1, unfrozen	DINOv3	NO	RT-DETR	2	Batch only (16 -> 2)	0.00000	0 %	0 %	cancelled

The fourth run is the closest to the author's intent: it respects his `freeze_backbone: false`, his learning rate and his 50 epochs. Only the batch was reduced, because without freezing it does not fit in 12 GB.

And it is the worst performer: 23 epochs at zero, against the 0.00055 of the same model with the backbone frozen. The run was cancelled once it was clear no improvement was coming.

Frozen vs. trainable backbone

Same model, same epoch 23:

	Frozen	Trainable
Loss per batch	0.17	0.66
mAP	0.00034	0.00000

Without freezing, the detector starts from random initialisation and sends chaotic corrections backwards that degrade what DINOv3 had already learned.

By class (experiment 1, the best one). Performance tracks colony size:

Class	Size	Precision	Recall	mAP of its head
mold_bact	22.8 px	3.35 %	7.47 %	0.00323
e_coli_bact	10.5 px	1.44 %	2.76 %	0.00047
bacillus_bact	9.2 px	0.75 %	1.16 %	0.00009
s_aureus_bact	7.1 px	0.51 %	1.18 %	0.00005
yeast_bact	7.1 px	0.43 %	0.54 %	0.00323

listeria_o_bact	7.1 px	0.22 %	0.40 %	0.00011
coliform_bact	6.8 px	0.43 %	1.05 %	0.00047
listeria_bact	5.7 px	0.78 %	1.38 %	0.00020
listeria_mi_bact	5.0 px	0.88 %	1.85 %	0.00011
listeria_m_bact	5.0 px	0.56 %	0.99 %	0.00011
entero_bact	4.3 px	0.36 %	0.72 %	0.00006
interferent	4.3 px	0.32 %	0.19 %	0.00005
tc_bact	3.9 px	0.27 %	0.48 %	0.00023

Conclusion

The model cannot be used to count colonies. Out of every 1,000 boxes it predicts, roughly 5 land on a real colony.

The problem is that it never tightens the boxes. On a plate with 50 colonies, out of 100 detections **28 land on top of a real colony**, but none of them fit closely enough: the best one reached an overlap of 0.485 where 0.5 is required.

Beyond the detector, the active learning loop is not implemented: the formula that decides which plates to send for human review exists, but nothing computes the inputs that formula needs. If it is run as is, every plate scores zero and the selection is arbitrary, without raising any error.

Inference evidence

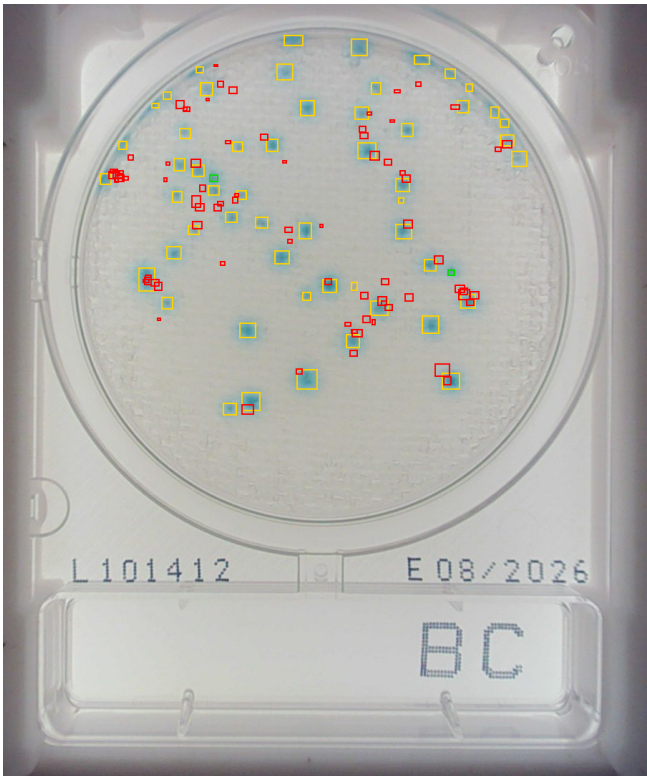
Five validation plates, one per head, run through the experiment 1 checkpoint (outputs/rtdetr_dinov3/model.pt). Each plate is shown in full and zoomed into its centre, where the boxes are actually legible.

Colour key (same in every image):

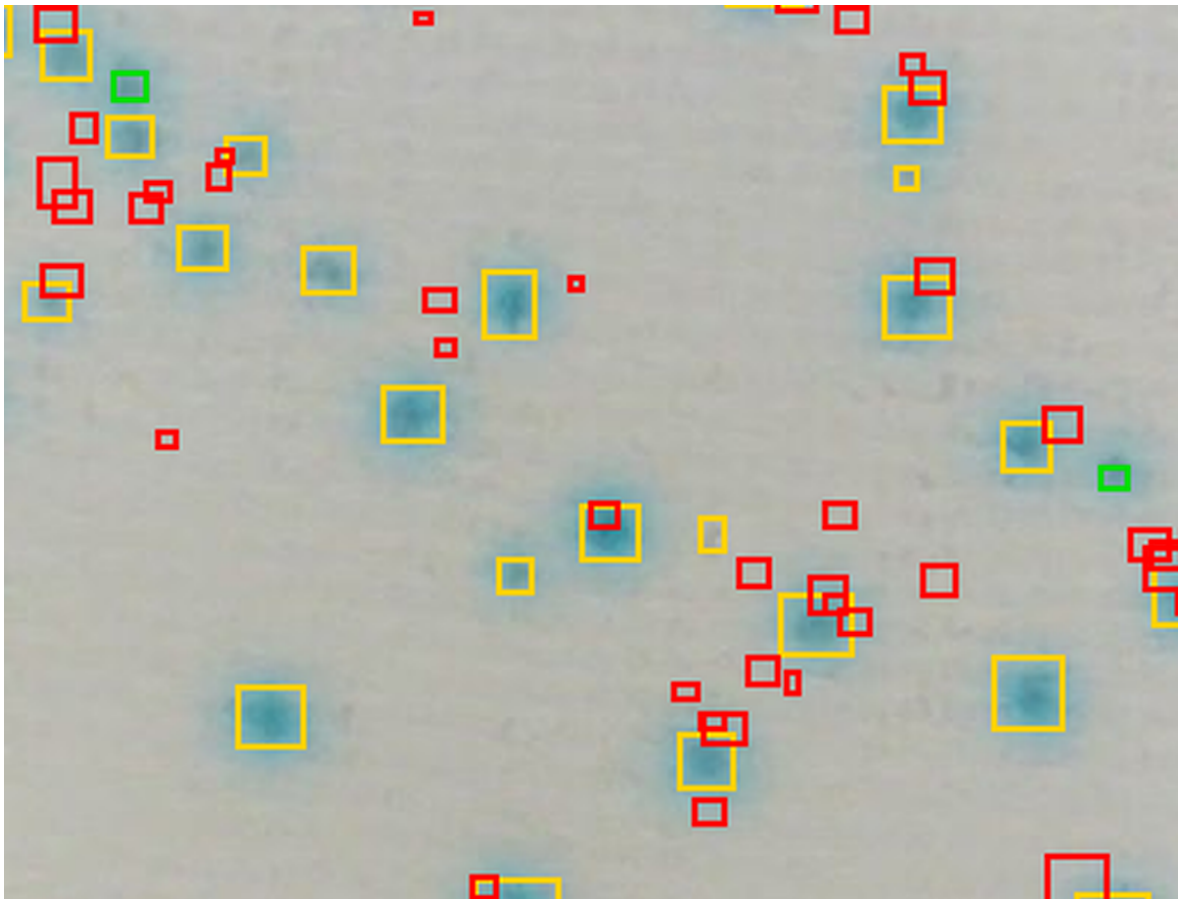
Colour	Meaning
GREEN	Hit - the model correctly detected a real colony (overlap ≥ 0.5)
RED	False prediction - a box where there is no colony, or too far off to count
YELLOW	Miss - a real colony the model did not detect

Plate BC — BC_Apracom_plate_bottom_002

Real colonies	Hits (green)	False predictions (red)	Missed (yellow)	Recall
56	2	77	54	3.6 %



Full plate.



Centre of the same plate, enlarged. Colonies measure only a few pixels in the original image, so the fit of each box is only visible at this zoom.

Plate ETB — ETB_Apracom_plate_bottom_038

Real colonies	Hits (green)	False predictions (red)	Missed (yellow)	Recall
28	0	33	28	0.0 %



Full plate.



Centre of the same plate, enlarged. Colonies measure only a few pixels in the original image, so the fit of each box is only visible at this zoom.

Plate LM — LM_Apracom_plate_bottom_090

Real colonies	Hits (green)	False predictions (red)	Missed (yellow)	Recall
24	1	27	23	4.2 %



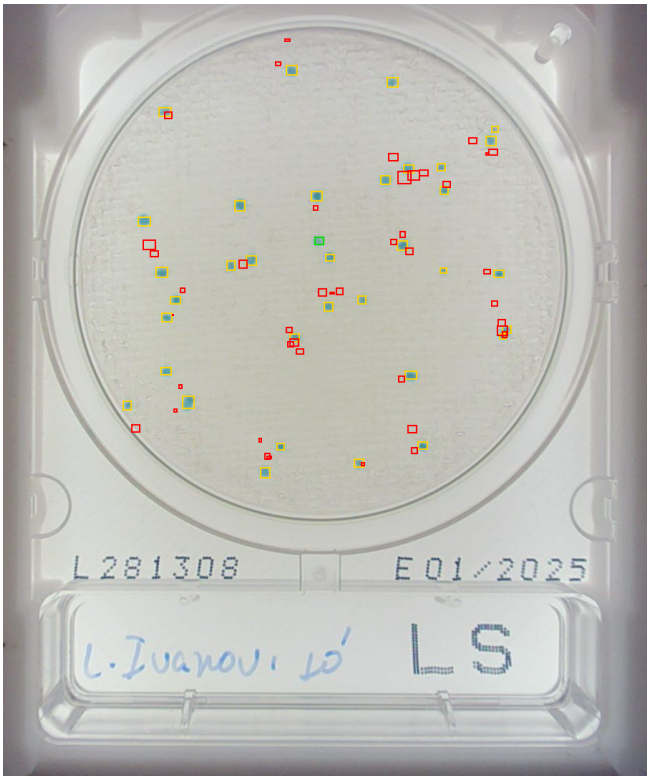
Full plate.



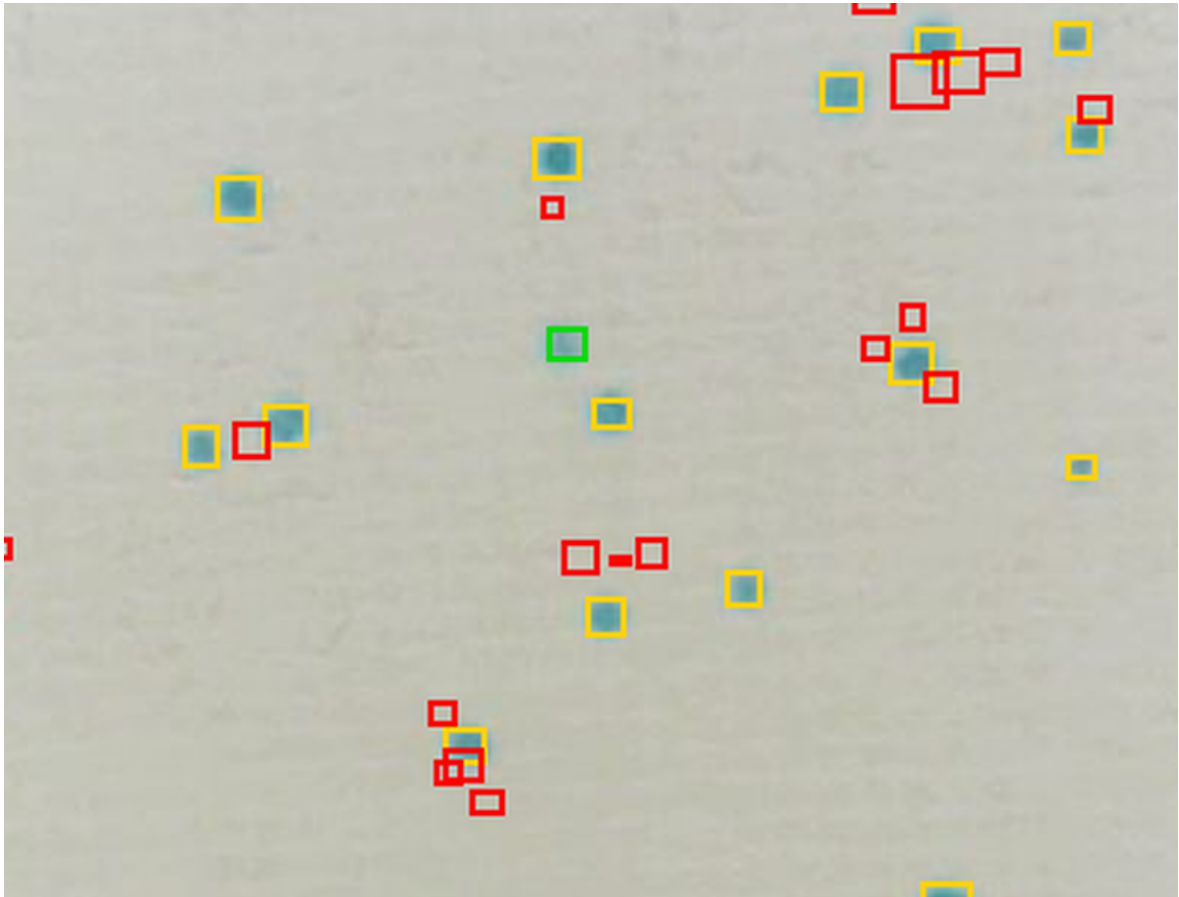
Centre of the same plate, enlarged. Colonies measure only a few pixels in the original image, so the fit of each box is only visible at this zoom.

Plate LS — LS_Apracom_plate_bottom_047

Real colonies	Hits (green)	False predictions (red)	Missed (yellow)	Recall
35	1	42	34	2.9 %



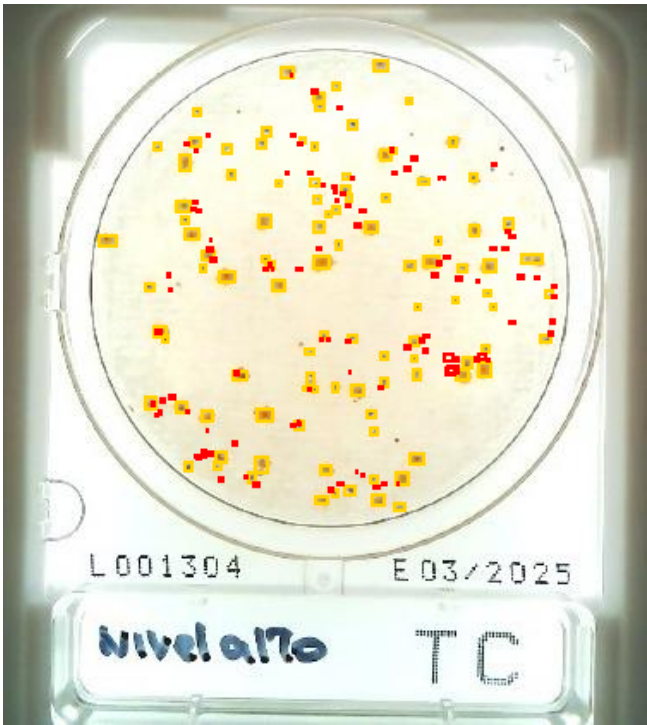
Full plate.



Centre of the same plate, enlarged. Colonies measure only a few pixels in the original image, so the fit of each box is only visible at this zoom.

Plate TC — TC(R)_Apracom_plate_bottom_112

Real colonies	Hits (green)	False predictions (red)	Missed (yellow)	Recall
102	0	100	102	0.0 %



Full plate.



Centre of the same plate, enlarged. Colonies measure only a few pixels in the original image, so the fit of each box is only visible at this zoom.